24.05.2025 (E)

SOMAIYA VIDYAVIHAR UNIVERSITY

Semester: Jan 2025 – April 2025		
Maximum Marks: 100	**Examination: ESE Examination**	**Duration:3 Hrs.**
Programme code: 01 **Programme: B.Tech. Computer Engineering**	**Class: TY**	**Semester:V(SVU 2020)**
Institute/School/Department: K. J. Somaiya School of Engineering	**Name of the department:** COMP	
Course Code: 116U01E514	**Name of the Course: Soft Computing**	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory **3) Assume suitable data wherever necessary**		

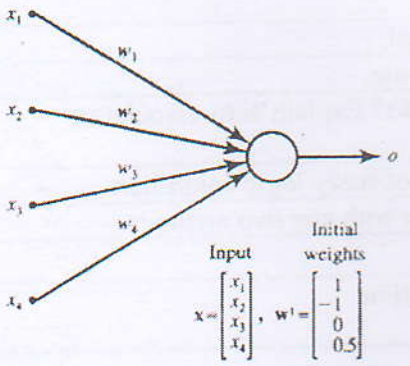
Que. No.	Question	Max. Marks
Q1	Solve any **Four**	**20**
i)	Differentiate between hard computing and soft computing with suitable examples.	5
ii)	Explain McCulloch-Pitts neuron model.	5
iii)	Write a short note on Delta learning rule.	5
iv)	What are associative memory networks? Explain auto-associative memory.	5
v)	Explain the concept and applications of fuzzy logic controllers.	5
vi)	Discuss the concept of defuzzification with any two methods.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	**10**
i)	Implement AND function using McCulloch-Fitts neuron (take binary data).	5
ii)	Explain Single layer and Multilayer Neural Network.	5
	OR	
Q2 A	Draw and explain architecture of Kohonen's self-organization feature map KSOFM.	10
Q 2 B	Solve any **One**	**10**
i)	Deign Adaline network for the following input and output x1 x2 t 1 1 1 1 -1 1 -1 1 1 -1 -1 -1 Show 1 Epoch.	10
ii)	Implement AND function using perceptron networks for bipolar inputs and targets. (Show 1 Epoch)	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	**20**
i)	Train' a hetero-associative memory network using Hebb rule. Store input row vector s =(s1,s2,s3,s4) to the output row vector t = (t1 t2). The vector pairs are given in the following table	10

Input Targets	s1	s2	s3	s4	t1	t2
1st	1	0	1	0	1	0
2nd	1	0	0	1	1	0
3rd	1	1	0	0	0	1
4th	0	0	1	1	0	1

ii)	Explain Error Back Propagation Training algorithm (EBPTA) with a help of diagram.	10
iii)	Apply Hebbian learning with continuous activation function on a very simple network as shown below:	



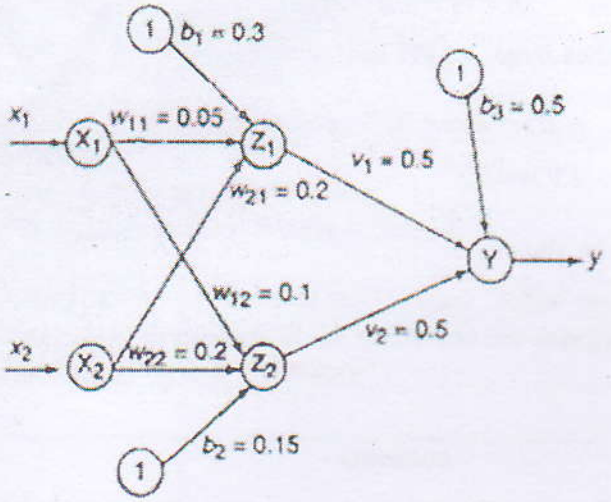
The network needs to be trained using the set of three input vectors as given below:

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ -2 \\ 1.5 \\ 0 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 1 \\ -0.5 \\ -2 \\ -1.5 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 1.5 \end{bmatrix}$$

for an arbitrary choice of learning constant c = 1.

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	**20**
i)	Deign Madaline network for the following inputs and outputs.	10

x_1	x_2	1	t
1	1	1	−1
1	−1	1	1
−1	1	1	1
−1	−1	1	−1

	Learning rate=0.05, Also assume the initial weight given in the following diagram.  Show 1 Epoch.	
ii)	Discuss the architecture of ART neural network.	10
iii)	Explain the steps involved in designing a fuzzy rule-based control system.	10

Que. No.	Question	Max. Marks
Q5	Solve any **four**	**20**
i)	What is supervised learning and how is it different from unsupervised learning?	5
ii)	Compare Hebbian and Winner-Take-All learning rules.	5
iii)	Explain the working of a Recurrent Neural Network	5
iv)	Describe membership functions and their types.	5
v)	Define fuzzy propositions and implications.	5
vi)	Explain Cognitron Neural Network	5